SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1001	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	Total Marks:	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	Lingeshwar T	Reg. No.:	192521226
Date:	01/10/2026	Duration:	60 minutes

Code of Conduct

I Lingeshwar T(192521226) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____Lingeshwar T_____

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted one problem statement. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. Understand the Problem Statement
    - Explain the problem in your own words.
    - Identify the objectives of the proposed system.
    - State the scope of the problem.
2. Identify Key Stakeholders and Challenges
    - Identify all the stakeholders involved in the system.
    - Explain the roles and responsibilities of each stakeholder.
    - Discuss the major challenges and issues faced by the stakeholders.

3. Analyze the Current Approach
    - Describe the existing system or current process.
    - Identify its strengths and limitations.
    - Analyze the problems or gaps that need to be addressed.
4. Propose a Suitable Solution Using Software Engineering Principles
    - Design a software-based solution for the given problem.
    - Describe the major functional modules of the proposed system.
    - Identify the functional and non-functional requirements.
    - Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.
5. Justify the Chosen Methodology/Framework
    - Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).
    - Justify why the selected methodology/framework is suitable for the proposed system.
    - Explain how it supports successful project development and maintenance.
6. Discuss Expected Outcomes and Limitations
    - Describe the expected benefits of the proposed solution.
    - Explain how the solution addresses the identified challenges.
    - Discuss potential risks, implementation challenges, and limitations.
    - Suggest possible future enhancements.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.

	(modularity, scalability, maintainability, security, etc.).			
Methodology/ Framework Justification (10%)	Selects the most suitable SDLC/model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	Mark
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25

Problem Statement

Many colleges currently manage student attendance and academic performance using manual registers, spreadsheets, or separate systems. This can lead to inaccurate records, data-entry errors, difficulty in tracking student performance, and delays in accessing information.
The proposed College Student Attendance and Academic Performance Management System is a web-based software system that digitally manages student attendance and academic records. Faculty members can record attendance and marks, students can view their attendance and performance, and administrators can manage the overall system.

1.Understand the Problem Statement

Problem Explanation:
The existing manual method of managing attendance and academic performance requires considerable time and effort from faculty members. Students may also find it difficult to know their current attendance percentage and academic performance.
The proposed system provides a centralized platform where attendance and academic information can be stored, updated, and accessed easily.

Objectives:

1. To maintain student attendance records digitally.
2. To record and manage students' academic marks.
3. To allow students to view their attendance and performance.
4. To identify students with low attendance.
5. To reduce manual work and data-entry errors.
6. To provide quick and organized access to academic information.
7. To improve communication between students, faculty, and administrators.

Scope:

The system will cover student management, attendance management, marks management, student dashboards, attendance alerts, administrator management, report generation, and user security.

Students can view their attendance and academic performance. Faculty members can record attendance and marks. Administrators can manage students, faculty, subjects, and system records.

---

2. Identify Key Stakeholders and Challenges

Stakeholders

1. Students

Students use the system to view their attendance, marks, subjects, and academic performance.

2. Faculty Members

Faculty members record student attendance, update marks, and monitor student performance.

3. College Administrators

Administrators manage student accounts, faculty accounts, subjects, and overall system information.

4. Department Management

Department management can use reports to monitor attendance and academic performance across students and subjects.

Roles and Responsibilities

Students are responsible for checking their attendance and academic records.

Faculty members are responsible for entering accurate attendance and marks.

Administrators are responsible for maintaining user accounts and system data.

Department management uses generated reports to monitor academic activities.

Major Challenges

1. Maintaining large numbers of student records.
2. Manual data entry can cause errors.
3. Attendance records may be difficult to track.
4. Students may not know their updated attendance percentage.
5. Generating reports manually takes time.
6. Unauthorized users may access confidential academic information.
7. Maintaining accurate and updated records can be difficult.

---

1. Analyze the Current Approach

Existing System

In the existing approach, colleges may use attendance registers, spreadsheets, or different systems to maintain student attendance and academic records.

Faculty members manually enter attendance and marks. Students may need to contact faculty or check separate records to know their academic status.

Strengths of the Current Approach

1. Simple to use.
2. Does not require complex software.
3. Can be implemented with existing college resources.
4. Faculty members are familiar with manual processes.

Limitations

1. Requires more manual work.
2. Data-entry errors can occur.
3. Records can be difficult to search.
4. Generating reports takes more time.
5. Students may not have immediate access to their records.
6. Maintaining large amounts of data is difficult.
7. Manual records may be damaged or lost.

Problems and Gaps

The current approach does not provide a centralized and convenient method for managing attendance and academic performance. There is a need for a system that provides quick access to updated records, reduces manual work, improves accuracy, and generates reports automatically.

---

4. Proposed Solution Using Software Engineering Principles

Proposed Solution

A web-based College Student Attendance and Academic Performance Management System will be developed to provide a centralized platform for managing attendance and academic records. The system will have separate access for students, faculty members, and administrators.

Major Functional Modules

1. Login and Authentication Module

Users can securely log in according to their roles.

2. Student Management Module

Administrators can add, update, view, and manage student information.

3. Faculty Management Module

Administrators can manage faculty information and assign subjects.

4. Attendance Management Module

Faculty members can record and update attendance for students.

5. Marks Management Module

Faculty members can enter and update internal and academic marks.

6. Student Dashboard

Students can view attendance, marks, subjects, and academic performance.

7. Attendance Alert Module

The system can identify students with low attendance and display alerts.

8. Report Generation Module

The system can generate attendance and academic performance reports.

Functional Requirements

1. The system shall provide user login.
2. The system shall allow administrators to manage students and faculty.

3. The system shall allow faculty to record attendance.
4. The system shall allow faculty to enter marks.
5. The system shall calculate and display attendance percentages.
6. The system shall allow students to view their academic records.
7. The system shall generate reports.
8. The system shall provide low-attendance alerts.

Non-Functional Requirements

1. Security: Student and academic data should be protected.
2. Usability: The system should be simple and easy to use.
3. Reliability: The system should provide accurate records.
4. Scalability: The system should support increasing numbers of students and records.
5. Maintainability: The software should be easy to update and maintain.
6. Performance: The system should respond quickly to user requests.

Software Engineering Principles

Modularity

The system will be divided into separate modules such as attendance, marks, student management, and authentication.

Scalability

The system can be expanded to support more students, departments, subjects, and features.

Maintainability

Separate modules make it easier to identify problems and update individual features.

Reliability

Proper validation and centralized data storage can help maintain accurate records.

Usability

A simple dashboard and clear navigation will make the system easy for students and faculty to use.

Security

Role-based access and authentication will restrict users from accessing unauthorized information.

---

5. Justify the Chosen Methodology/Framework

Selected Methodology: Agile Scrum

The Agile Scrum methodology is suitable for developing the College Student Attendance and Academic Performance Management System.

The system can be divided into smaller modules such as login, student management, attendance, marks, reports, and notifications. Each module can be developed and tested in separate iterations.

Why Agile Scrum is Suitable

1. Development can be divided into small sprints.
2. Each module can be tested before moving to the next module.
3. Changes in requirements can be handled easily.
4. Regular feedback can be obtained from students and faculty.
5. Errors can be identified and corrected during development.
6. New features can be added in future iterations.
7. Continuous testing and improvement can be performed.

Example Sprint Plan

Sprint 1: Login and user authentication.

Sprint 2: Student and faculty management.

Sprint 3: Attendance management.

Sprint 4: Marks and academic performance.

Sprint 5: Reports and attendance alerts.

Sprint 6: Testing, security, and final deployment.

Agile Scrum therefore supports continuous development, testing, feedback, and maintenance of the proposed system.

---

6. Expected Outcomes and Limitations

Expected Outcomes

1. Reduced manual work for faculty members.
2. Improved accuracy of attendance records.
3. Easy access to student academic information.
4. Faster generation of attendance and academic reports.
5. Students can monitor their attendance regularly.
6. Low-attendance students can be identified quickly.
7. Improved organization of academic records.
8. Secure access to student information.

How the Solution Addresses the Challenges

The proposed system replaces many manual processes with a centralized digital system. Attendance and marks can be stored electronically, reducing the need for paper records. Students can access their information through their dashboard, while faculty members can update records directly. Automatic calculations and reports can reduce errors and save time.

Potential Risks

1. Unauthorized access to student information.
2. Incorrect data entered by users.
3. System failure or technical problems.
4. Data loss if proper backup is not maintained.
5. Users may require training to use the system.

Implementation Challenges

1. Migrating existing student records into the new system.
2. Training faculty and administrators.
3. Maintaining data security.
4. Ensuring reliable system availability.
5. Integrating the system with existing college infrastructure.

Limitations

1. The system requires a computer or mobile device and network access.
2. Incorrect information entered by faculty may affect reports.
3. System maintenance is required.
4. Initial implementation may require time and resources.
5. The system depends on proper database and server availability.

Future Enhancements

1. Mobile application for students and faculty.
2. Parent notification system.
3. Biometric or facial-recognition attendance.
4. Email and SMS notifications.
5. AI-based academic performance prediction.
6. Integration with online examination systems.
7. Cloud-based deployment.
8. Advanced analytics and performance dashboards.

Conclusion

The College Student Attendance and Academic Performance Management System provides a centralized software solution for managing student attendance and academic performance. It reduces manual work, improves accessibility, supports accurate record management, and provides useful reports. Using Agile Scrum allows the system to be developed incrementally with continuous testing, feedback, and improvement.